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# Life cycle analysis of greenhouse gas emissions for fluorescent lamps in mainland China

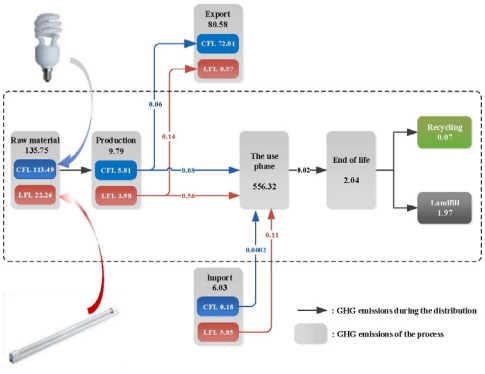
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## HIGHLIGHTS

- GHG emissions of compact with linear fluorescent lamps were compared.
- GHG emissions of fluorescent lamps at the national level of China were qualified.
- More than half of GHG emissions were embodied in the exported fluorescent lamps.

## GRAPHICAL ABSTRACT




Greenhouse gas emissions associated with fluorescent lamps in mainland China in 2011 (million tons $CO_2$-equivalents)

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## ABSTRACT

China is the world's largest emitter of carbon dioxide, and it is also one of the largest fluorescent lamp consuming and producing country in the world. However, there are few studies evaluating greenhouse gas (GHG) emissions of fluorescent lamps in China. This analysis compared GHG emissions of compact fluorescent lamps with linear fluorescent lamps using life cycle assessment method in China's national conditions. The GHG emissions of fluorescent lamps from their manufacture to the final disposal phase on the national level of China were also quantified. The results indicate that the use phase dominates the GHG emissions for both lamps. Linear fluorescent lamp is a better source of light compared to compact fluorescent lamp with respect to GHG emissions. The analysis found that in 2011, China generated around 710.90 million tons $CO_2$-eq associated with fluorescent lamps. The raw material production and use phases accounted for major GHG emissions. More than half of GHG emissions during the domestic production were embodied in the exported lamps in recent years. This urges the government to take necessary measures that lead to more environmental friendly production, consumption and trade patterns.
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## 1. Introduction

Climate change has been an urgent and serious global problem, and it is mainly caused by the greenhouse gas (GHG), especially carbon

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dioxide ($CO_2$). As a result of the associated greenhouse effect, the global temperature increased by 0.85 °C on average from 1880 to 2012 (Zhang and Wang, 2015). At the Paris climate conference in December 2015, 195 countries adopted the first-ever universal, legally binding global climate deal (European Commission, 2015). The agreement sets out a global action plan to put the world on track to avoid dangerous climate change by limiting global warming to well below 2 °C. It is estimated that lighting consumes about 19% of the global electricity production, of which about 62% is consumed by fluorescent lamps (FLs) (IEA, 2006). Electricity production can generate tremendous GHG mainly due to fossil fuel combustion, therefore the amount of GHG emissions associated with FLs is considerable. In addition, with the phasing out of incandescent lamps in several countries, such as China (NDRC, 2011), America, the European Union, Australia (Ramroth, 2008; Waide, 2009) and so on, the popularity of FLs is rising. So it is increasingly necessary to investigate GHG emissions related to FLs.

In 2007, China surpassed the U.S. as the largest emitter of carbon dioxide in the world (Gregg et al., 2008), mainly because of rapid economic development and high industrial energy consumption. Global carbon dioxide emissions from fossil fuel combustion and industrial processes (cement and metal production) increased to 35.3 billion tons in 2013, out of which 10.3 billion tons came from China, accounting for 29% (Olivier et al., 2014). China has been facing increasing international pressure to reduce carbon emissions. It is urgent and vital to accelerate the pace of GHG emission reduction in China to the success of global efforts in addressing climate change.

In 2009, China announced the goal of controlling GHG emissions, i.e. carbon dioxide emission intensity in 2020 decreasing by 40%–45% than that in 2005 (Xinhuashe, 2009). China in 2014, for the first time proposed that it intended to achieve the peaking of $CO_2$ emissions around 2030 and to make the best efforts to peak early and intended to increase the share of non-fossil fuels in primary energy consumption to around 20% by 2030. China also proposed that by 2030, carbon dioxide emissions per unit of GDP reduced by 60%–65% compared with that in 2005. China's commitment related to carbon emissions will bring huge pressure to the transformation of domestic energy structure and industrial restructuring. In addition, China is one of the largest FL consuming and producing countries in the world. In 2011, domestic production of compact fluorescent lamps (CFLs) in China accounted for over 80% of the world's production (Xin et al., 2012). There is an increasing need to analyze GHG emissions of FLs in China.

A few of life cycle assessment (LCA) studies have been performed analyzing environmental impacts of FLs (Bakri et al., 2010; Department for Environment, Food and Rural Affairs, DEFRA, 2009; OSRAM, 2009; Principi and Fioretti, 2014; Quirk, 2009; Tahkamo et al., 2014; Tan et al., 2015; DOE, 2012; Welz et al., 2011). These studies can be divided into two categories: one focused on the comparative of FLs with other luminaires, such as light-emitting diodes (LEDs) and incandescent lamps; the other analyzed environmental performance of FLs. For the first category, Quirk (2009) investigated the full life-cycle costs and benefits of using a 13 W CFL and a 6 W LED to compared with a less efficient 60 W incandescent lamp. The study found that CFL and LED lamp are roughly four times more efficient than incandescent lamps. The OSRAM (2009) evaluated the environmental impacts of a 40 W incandescent, 8 W CFL and 8 W LED lamp. The study concluded that current LED lamps, as of 2009, are comparable to CFLs of life-cycle energy, and both provide significant energy savings compared to incandescent lamps. The previous estimates have found that fluorescent lamps result in a clearly lower environmental impact than incandescent lamps. The use phase is the main contributor to GHG emissions. However, most previous studies were not evaluating GHG emissions of FLs in China's national conditions. For the second category, Bakri et al. (2010) compared the environmental performance of electromagnetic and electronic ballast used for FLs. Tan et al. (2015) evaluated environmental impacts of linear fluorescent lamps (LFLs) and compact fluorescent lamps (CFLs) based on Chinese conditions, yet the GHG emissions are not the key point. In addition, former studies focus solely on the product level, no study has been conducted at the national level.

There are a number of studies regarding the role that Chinese trade plays in global $CO_2$ emissions (Liu et al., 2015a,b; Su and Ang, 2013; Xie et al., 2015; Yunfeng and Laike, 2010; Zhang et al., 2014). According to Su and Ang (2013), from 2002 to 2007, the carbon emissions embodied in China's exports increased from 764 to 1775 Mt. Xie et al. (2015) implemented a multi-regional input-output framework to evaluate the carbon footprints and embodied carbon flows for eight regions of China. It was found that the construction, electricity/stream supply, and machine manufacturing rank as the top sectors with the largest total carbon emissions. Liu et al. (2015b) found that in China, the transferred carbon emissions embodied in demand–supply chains driven by consumption and export both showed rapid growth during 1997–2007. Above all, input–output analysis is the most common methodology used in these literatures to identify effects of production and estimate the embodied emissions in international trade. Few studies use LCA, the so-called "bottom up" approach. However, LCA has advantages over input–output analysis in the embodied carbon evaluation of single products because it collects production information from the technical details (Xie et al., 2015), and it can get relatively accurate results. Most of the studies focused on the whole nation emissions of China, such as agriculture, equipment manufacturing, however, the role of FLs was small and vague. Furthermore, details of FLs regarding their production, import, export, and waste management are missing.

This study consists of two parts: one part is to compare GHG emissions of CFLs with LFLs in China's national conditions, and all the materials, energy use and pollutant emissions to the environment from each process are analyzed; the other part is to quantify GHG emissions of FLs from their manufacture to the final disposal phase on the national level of China. The amount of GHG emissions related to the exported FLs is provided as well. The goal is to determine the life cycle stages with material and energy inputs that cause the greatest impacts and allow a better understanding of the GHG emission characteristics of FLs. The analysis also helps government understand GHG emissions embodied in exported products and provides a reference to policy makers for the GHG emission reduction.

## 2. Methodology

Life cycle assessment is scientific methodology that enables researchers to quantify and provide a comprehensive view of the environmental and sustainability impacts of a product across a range of categories for a product over its entire life cycle. The LCA is conducted according to the international standards ISO 14040 (ISO, 2006a) and ISO 14044 (ISO, 2006b). EBalance, which is Chinese LCA software, has been used to model different processes in the life cycle of the two lighting technologies. EcoInvent and Chinese Life Cycle Database (CLCD) are used as databases. CLCD is a database based on Chinese conditions, and the data represents the average level of China's manufacturing technology and market. GHG are converted to the measurement of carbon dioxide equivalents ($CO_2$-eq), using the Intergovernmental Panel on Climate Change (IPCC) 100-year Global Warming Potential (GWP) factors. A cradle to grave approach has been taken up for the study in order to include the raw material production, manufacturing, distribution, use, and end-of-life stages of the lamps.

### 2.1. Comparison of compact fluorescent lamps with linear fluorescent lamps

The life cycle of the stages is shown in Fig. 1. The lamps selected were a 15 W CFL and a 28 W T5 LFL. Functional unit selected for this study was 62 million lumen–hours of lighting service, which is approximately representative of total light output of the 28 W T5 LFL over its lifetime. And it is equal to 8 times of the CFL to provide light throughout its stated life.

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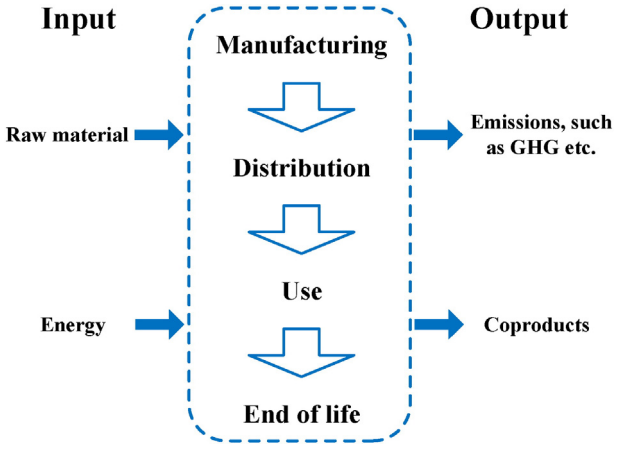



**Fig. 1.** Life-cycle stages of lamps.

*2.1.1. Raw material production*

This stage accounts for the resource use and emissions associated with the production of the various raw materials that go into the final product and their transportation to the manufacturing place. The data for the inventory analysis were collected from literature, EcoInvent and CLCD databases. Table 1 presents the materials used in manufacturing the lamp and ballast for the CFL. The T5 LFL consists of a large glass tube with aluminum end caps, as well as electronic ballast and so on, and Tables 2–3 provide the list of all the raw materials that are assumed to be used in the LFL.

Most of the FL manufacturers have located in the southeast provinces, such as Zhejiang, Guangdong, Jiangsu, and Fujian, accounting for more than 87% of the production in China (Xin et al., 2012), as in Fig. 2. Therefore, the electricity generation during the production was chosen as the mix of Chinese southeast.

**Table 1**
Raw material inventory for the CFL.
Data source: US Department of Energy (DOE) (2012).

| Material | Database | Amount |
|---|---|---|
| Argon gas | CLCD | 0.004 g |
| Nitrogen gas | CLCD | 0.119 g |
| Oxygen gas | CLCD | 0.159 g |
| Hydrogen gas | CLCD | 0.002 g |
| Neon gas | EcoInvent | 0.0004 g |
| Ammonia | CLCD | 0.13 g |
| Noble earths | EcoInvent | 0.001 g |
| Yttrium oxide | EcoInvent | 1.37 g |
| Nitric acid | CLCD | 7.9 g |
| Sulfuric acid | CLCD | 1.67 g |
| Aluminum oxide | CLCD | 0.008 g |
| Lead | CLCD | 0.19 g |
| Copper | CLCD | 0.402 g |
| Nickel | CLCD | 0.003 g |
| Brass | EcoInvent | 1.65 g |
| Cast iron | EcoInvent | 0.029 g |
| Chromium | EcoInvent | 0.0002 g |
| Mercury | CLCD | 0.004 g |
| Capacitor | EcoInvent | 40 pcs. |
| Coil miniature | EcoInvent | 3 pcs. |
| Diode SMD | EcoInvent | 40 pcs. |
| PWB | EcoInvent | 3.7 g |
| Resistor SMD | EcoInvent | 40 pcs. |
| Thermistor, NTC | EcoInvent | 0.19 g |
| Transistor power large | EcoInvent | 3.70 g |
| Resin glue | EcoInvent | 4.5 g |
| Solder paste | EcoInvent | 0.3 g |
| Glass tube | EcoInvent | 1.20 g |
| Housing top & bottom (PBTP) | EcoInvent | 2.39 g |
| Natural gas | CLCD | 10.7 kg |
| Power | CLCD | 3.13 MJ |
| Packaging | EcoInvent | 81 g |

**Table 2**
Raw material inventory for the LFL.
Data source: Department for Environment, Food and Rural Affairs (DEFRA) (2009).

| Material | Database | Amount |
|---|---|---|
| Argon gas | CLCD | 2.1 g |
| Krypton gas | EcoInvent | 2.1 g |
| Mercury | CLCD | 0.01 g |
| Steel | EcoInvent | 2.325 g |
| Caps | CLCD | 6.3 g |
| Wiring | CLCD | 0.557 g |
| Sockets | CLCD | 5 g |
| Coil | CLCD | 43.4 g |
| Metal film capacitor | CLCD | 4.65 g |
| ELKO component | CLCD | 3.1 g |
| Luster terminal | CLCD | 2.325 g |
| Phosphor | EcoInvent | 4.2 g |
| Base | EcoInvent | 500 g |
| PCB | EcoInvent | 15.5 g |
| Housing | EcoInvent | 79.05 g |
| PET film | EcoInvent | 3.1 g |
| Solder pastes | EcoInvent | 1.55 g |
| Glass | CLCD | 194.29 g |
| Polycarbonate | EcoInvent | 1010 g |
| Packaging (plastic) | CLCD | 10 g |
| Packaging (card) | EcoInvent | 4.11 g |

*2.1.2. Manufacturing*

The manufacturing stage takes all of the raw materials defined above, as delivered to the place of production, and accounts for the energies used and emissions related with fabricating the final lamp. The manufacturing activities are based on the assembly of LCD screen.

*2.1.3. Distribution*

Distribution stage includes the transport from the manufacturer site to the place it is used. The estimate of transport is based on distance from the main producing areas of FLs, which are mainly located in the southeast region of China, such as Zhejiang, Guangdong, to the main use provinces of FLs, such as Shanghai, Beijing, and it is estimated to be 1800 km by lorry within China. In addition, packaging is taken into account. For the CFL, the lamp itself weighs 153 g and the card-stock packaging was measured at 81 g, taken together the lamp inside the box totals approximately 234 g (DOE, 2012). For the LFL, the lamp weighs about 1.88 kg and taken with the packaging is 1.895 kg (DEFRA, 2009).

*2.1.4. Use*

It is assumed that there was no diminishing performance during the lifetimes of lamps, which means that the power consumption and light emitted per watt were constant. It was also considered that the luminaire systems would be powered by average Chinese electricity.

**Table 3**
Labor to process raw material for the LFL.
Data source: Department for Environment, Food and Rural Affairs (DEFRA) (2009).

| Name | Database | Amount |
|---|---|---|
| Sheet rolling, aluminum | EcoInvent | 14.05 g |
| Steel product manufacturing, average metal working | EcoInvent | 500 g |
| Wire drawing, copper | EcoInvent | 0.56 g |
| Injection molding | EcoInvent | 1017.3 g |
| Sheet rolling, chromium steel | EcoInvent | 81.38 g |
| Extrusion, plastic film | EcoInvent | 13.1 g |
| Aluminum product manufacturing, average metal working | EcoInvent | 43.4 g |

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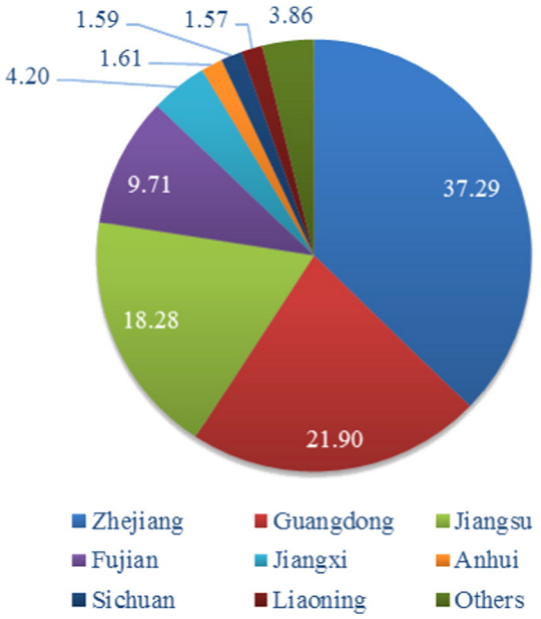



**Fig. 2.** The proportion of fluorescent lamp production for various provinces in 2011 (unit: %). Data source: Xin et al. (2012).

*2.1.5. End of life*

In the end-of-life stage detailed processes for disposal, recycling and landfill as municipal solid waste were taken into account. Waste management of spent FLs has not received sufficient attention in China. Most of spent FLs are discarded as the household waste. The available data on FLs recycling in China is absent. China is developing that the recycling rate was assumed and applied as 5% (Eckelman et al., 2008). The remaining FLs were assumed to be disposed to be landfilled. In addition to the lamps itself, the fate of its packaging was also considered. 80% of cardboard boxes were assumed to be recycled, the other 20% to be landfilled.

*2.2. 2.2 GHG emissions of FLs on the whole nation of China*

This part is to quantify GHG emissions of FLs from their manufacture to the final disposal phase on the national level of China. The time margin is the year 2011. Quantities of FLs in production, import and export in China were obtained from Tan and Li (2014). Fig. 3 is a simplified illustration of the boundaries and material flows for the system.

There are several different kinds of fluorescent lamps, including CFLs, LFLs and circular FLs. Considering the fact that the quantity of circular FLs are relatively small compared with LFLs, the circular lamps are grouped into the linear type. The amount of GHG emissions of FLs associated with production, import or export is based on the results of each lamp with the quantities of CFLs and LFLs in China. As mentioned above, the transport distance of FLs produced and used in China was estimated as 1800 km by lorry; for exported FLs, it was assumed as 900 km from the manufacturer site to the port in Shanghai. And it was assumed as 1100 km for the distribution within China regarding the imported FLs.

During the use phase, it was modeled on the basis of the energy consumption in the electricity generation. The GHG emissions from FLs in the use phase are formulated as Eq. (1).

$$E = T \cdot C \cdot R_1 \cdot R_2 \quad (1)$$

where E is the mass of GHG emissions during the use phase; T is total electricity generation; C is GHG emissions per kwh; $R_1$ is the ratio of lighting consuming in the total electricity consuming; $R_2$ is the ratio of FLs consuming in the total lighting consuming.

## 3. Results and discussion

*3.1. Comparison of CFLs with LFLs*

For the CFL, the total GHG emissions during its life are 145.892 kg $CO_2$-eq. Fig. 4 shows the impact of the CFL life cycle divided into raw material, manufacturing, transport, use and end of life. The largest contributor of GHG emissions is the use stage, which accounts for about 82.42% of the total emissions. The next most significant stage of the life cycle is the raw material production, representing 16.64%. Manufacturing is the third most impactful stage with an impact of approximately 0.85%. The remaining two life cycle stages – disposal and transport – constitute 0.06% and 0.03%, respectively. The impact associated with transport was found to be virtually negligible, even though the packaged CFLs travel about 1800 km from factory to consumers.

For the LFL, the GHG emissions of the life cycle are shown as Fig. 5. The total GHG emissions during its life are 685.773 kg $CO_2$-eq. The largest contributor to environmental impact is energy in use, which represents 98%. The second most significant impact is the raw materials used in manufacturing the LFL. These include a range of components, such as the fitting and ballast and so on. Manufacturing is the third most impactful stage in the life cycle, with just 1.38%. The disposal and transport impacts are extremely low, both are about 0.05%.

As shown in Fig. 6, the CFL has the larger GHG emissions, with 1.167 tons of emissions associated with the functional unit of 62 million

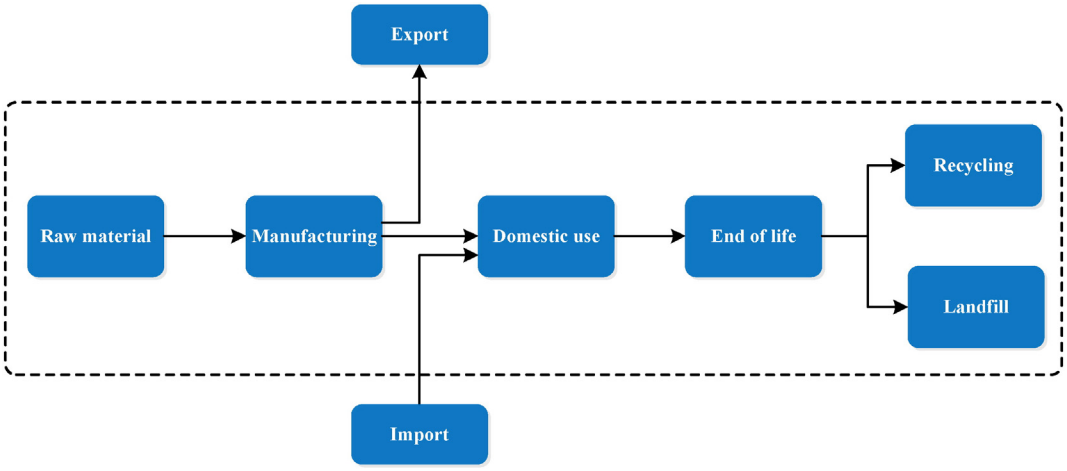



**Fig. 3.** The model of GHG emissions in China.

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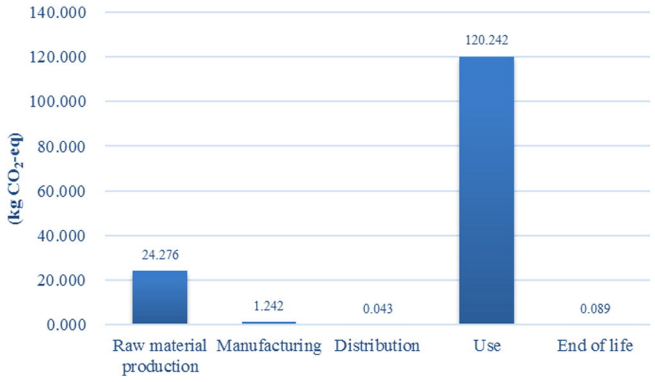



**Fig. 4.** GHG emissions for a CFL.

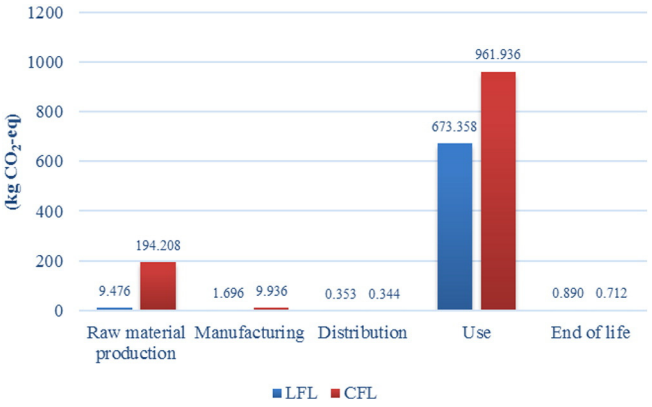



**Fig. 6.** Comparison of CFLs with LFL of the functional unit.

lumen-hours of light. The LFL represents a 41.25% reduction over the CFL for equivalent lighting service. The use stage is the dominating factor for both lamps, and the LFL would produce about 70.00% (673.385 kg $CO_2$-eq) of the GHG that would be emitted by the CFLs (961.936 kg $CO_2$-eq) during the use stage. The CFLs have lower impact than LFL in the distribution and end-of-life stage, but both of which have very little effect on the total life cycle impact due to the dominance of the use phase. Above all, the LFL is more environmental friendly than the CFL from the point of view of GHG emissions.

In this case a sensitivity analysis has been performed on the use phase of the alternative lighting technologies by changing the Chinese electricity mix to average European electricity mix. When the use is modeled by using the European mix, the GHG emissions during the use phase for a CFL are from 120.242 kg $CO_2$-eq to 70.850 kg $CO_2$-eq, decreasing by 41.10%. For the LFL, the emissions are from 673.358 kg $CO_2$-eq to 396.759 kg $CO_2$-eq, reducing by 41.08%. The electricity generation of China is dominated by thermal power, and it is almost all generated from coal combustion. Thermal power and hydropower contributes to approximately 81.3% and 14.8% of the electricity generation in China, respectively (NBS, 2012). However, the electricity in Europe describes an energy system where renewables and nuclear account for a large share of the generation. It indicates that with the change of cleaner electricity mix, the emissions of GHG decrease significantly.

### 3.2. GHG emissions of FLs at the national level of China

The analysis finds that in 2011, as illustrated in Fig. 7, China generated around 710.90 million tons $CO_2$-eq associated with FLs, accounting for about 8.16% of the total emissions in China. The proportion of impact attributable to energy in use was particularly high, which was about 78.26%. The next most significant stage was the raw material production, which constituted about 19.20%. Manufacturing was the third most significant stage with an impact of approximately 1.38%. The

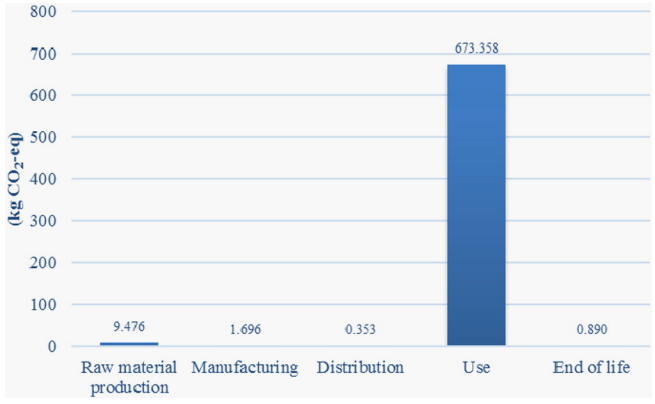



**Fig. 5.** GHG emissions for a LFL.

remaining two stages – disposal and distribution – accounted for 0.29% and 0.14%, respectively. During the distribution stage, the total GHG emissions were about 0.97 million tons $CO_2$-eq, of which about 0.20 million tons $CO_2$-eq were associated with exported lamps, accounting for 20.61%. It is obviously that the size of GHG emissions embodied in export was also significant. About 80.58 million tons GHG emissions were embodied in export, while the emissions in import were about 6.03 million tons.

On one hand, Fig. 8 shows that from 2005 to 2011, GHG emissions from domestic production increased 153%, from the 5744 million tons $CO_2$-eq in 2005 growth to the 14,554 million tons $CO_2$-eq in 2011. This is mainly due to the rapid growth of production of FLs in China, in 2005 the output of FLs was 2.876 billion units, and in 2011 reached to 7.024 billion. On the other hand, GHG emissions from raw material production were far greater than those from manufacturing. During the year 2005 to 2011, the contributor came from raw material production accounting for more than 70% of the total domestic production.

As Fig. 9 illustrates, GHG emissions related to CFLs in China is far greater than those of LFLs, such as in 2011, the former was about 4.5 times as the latter. This is because the quantity of CFL production was larger than the LFLs in recent years; the former was about 2 times as the latter. Another reason is that GHG emissions of a single CFL are more than those from the linear type. From 2005 to 2011, the GHG emissions of CFLs increased fast, but LFLs were not particularly evident. The production of CFLs increased from about 1.74 billion in 2005 up to 4.67 billion in 2011, increasing by 170%, while LFLs showed an increase of 112%, from 1.11 billion in 2005 to 2.35 billion in 2011.

GHG emissions from FLs of domestic use, export and import are presented in Fig. 10. It is clear that the emissions in domestic use of FLs were fewer than that of export, and more than half of GHG emissions from domestic production were embodied in exported lamps. Compared with GHG emissions embodied in export, the emissions in import were negligible. In the case of LFLs, the emissions in export were about equal to that in import. The emissions associated with LFLs in import were larger than that of CFLs. This is because the import volumes of LFLs were much larger than that of CFLs. The present method of estimating GHG emissions is within the national border in the United Nations Framework Convention on Climate Change (UNFCCC) and the Kyoto Protocol, and it is also called producer responsibility. The UNFCCC system boundary includes all GHG emissions and removals taking place within national (including administered) territories and offshore areas over which the country has jurisdiction, so it is named production-based emission accounting (Liu et al., 2015b). The emissions associated with exports are included but those associated with imports are excluded from the national account. This leads to some developed countries transferring high energy consumption and high emissions of production to other countries or import the corresponding products, thereby avoiding the clean production mechanism and reducing their reduction obligations of carbon emissions. It is estimated that more than half of

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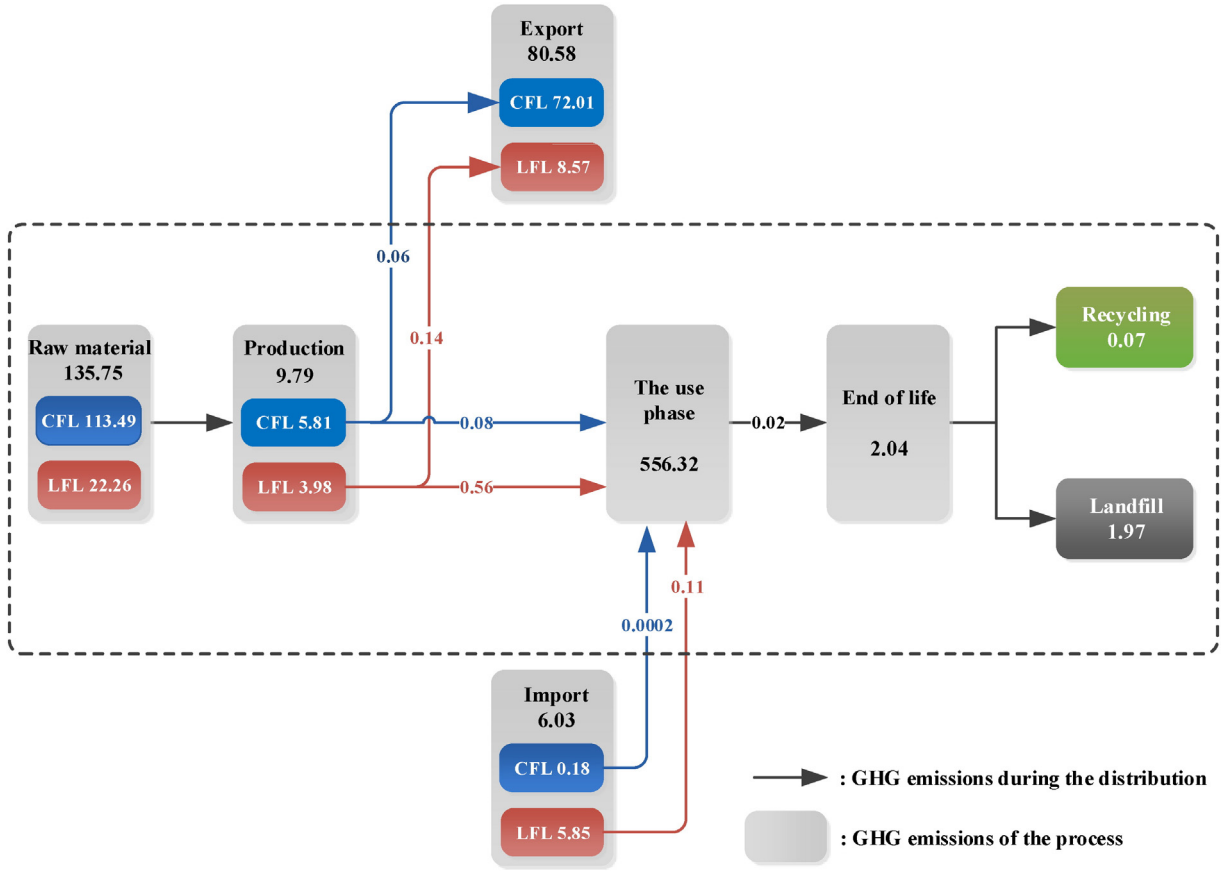



**Fig. 7.** GHG emissions associated with FLs in China in 2011 (unit: million tons $CO_2$-eq).

GHG emissions during the domestic production was embodied in the exported lamps in China. These emissions were attributed to foreign consumption and therefore are not only subject to the Chinese responsibility. Those who consume the lamps made in China should also share the responsibility. China as a "world factory", manufactures a large number of products for other countries. Carbon emissions embodied in these products cannot be ignored. China needs to clear up the hidden energy consumption and carbon emissions from the perspective of international trade, so as to reasonably define the responsibilities and obligations of reduction in the global GHG emissions.

The sensitivity analysis also has been conducted by changing the Chinese electricity mix to average European electricity mix on the use phase. If the use phase is employing the European mix, total GHG emissions are 327.79 million tons $CO_2$-eq, decreasing by 228.53 million tons. It is reducing significantly, having a reduction of about 30.27%. As mentioned above, the electricity generation, which is based on coal consumption, is of high-carbon scenario in China. It should be noted that it is very difficult to make a considerable change in coal-dominated energy consumption structure of China, and GHG emissions of China will continue to grow.

For the purpose of reducing GHG emissions, the share of electric power generated from renewable sources should be increased and clean energy developed, such as wind energy and solar energy. The efficiency of energy use can be improved, including the use of cleaner production and other technologies. In addition, it should be increasing the carbon uptake of terrestrial ecosystems, such as afforestation, forest land restoration. To a certain extent, it can alleviate the pressure of carbon emission reduction, so as to provide time for the process of domestic industrialization. It can be advocated to the public having low carbon consumption, such as encouraging people to travel by subway, bus and

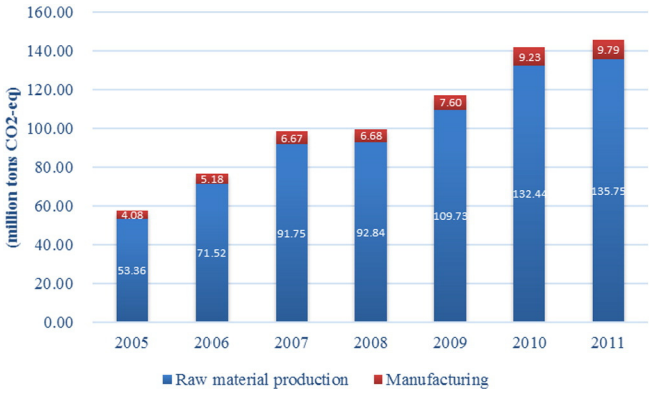



**Fig. 8.** GHG emissions from FLs of raw material production and manufacturing.

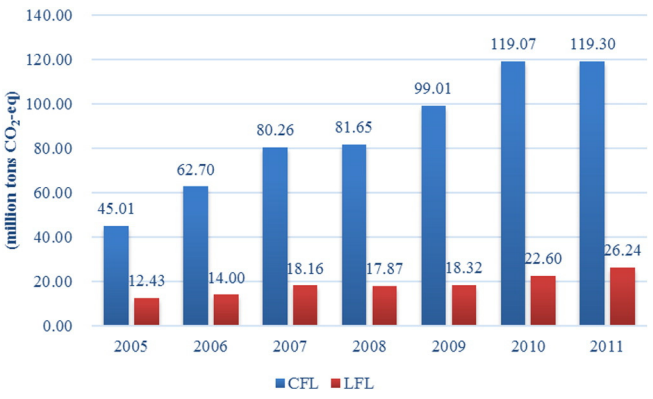



**Fig. 9.** GHG emissions from CFLs and LFLs.

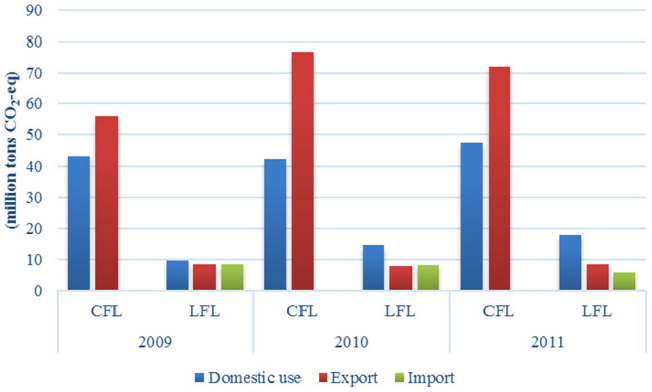



**Fig. 10.** GHG emissions from FLs of domestic use, export and import.

other modes of public transport, promoting the development of electric cars and other low emission vehicles.

It also should be minded that although the disposal stage has very little effect on the total life cycle impact due to the dominance of use and manufacturing stages, it is recommended to increase the recycling rate of spent lamps. This is mainly because FLs contain mercury, which is one of the most toxic elements in the periodic table. Mercury has adverse effects on the nervous system and its development, as well as on the cardiovascular system, immune system, and reproductive system, it also does harm to kidneys, especially for infants, children and pregnant women (UNEP, 2013). Waste management of end-of-life FLs has not received sufficient attention in China. Most areas of China are short of special recycling infrastructure and some citizens' environmental awareness is relatively weak.

## 4. Conclusions

This analysis compared GHG emissions of CFLs with LFLs in China's national conditions. GHG emissions of FLs from their manufacture to the final disposal phase at national level of China were also quantified. As for the comparison of CFLs with LFLs, the results indicated that the greatest negative impact of both kinds of lamps is in the use phase, when electricity for the running of the lamp is used. The GHG emissions from the distribution and disposal stages can be neglected for both lamps. LFL is a better source of the light than CFLs regarding the GHG emissions. The sensitivity analyses clearly indicated that with the change of cleaner electricity mix, the emissions decreased significantly.

In the case of GHG emissions on the national level of China, the analysis finds that in 2011, China generated around 710.90 million tons $CO_2$-eq associated with FLs. The raw material production and use phase accounted for the majority of the GHG emissions. More than half of GHG emissions during the production phase were embodied in the exported lamps in recent years. Those who consume the lamps made in China should also share the responsibility and it is reasonable to establish a new global framework to allocate emission responsibilities. It should be noted that though it is very difficult to make a considerable change in coal-dominated energy consumption structure of China, it needs great efforts to lower dependence of coal in the energy consumption. The government needs to improve the export structure through policies and measures. As a large exporting country, China produces tremendous exported goods with huge resource consumption and GHG emissions, so it is very necessary to figure out the GHG emission characteristics of other exported goods.

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